

Mapping Mecklenburg County: Exploring Census Data for Potential Communities of Interest



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Abstract Electoral districting remains a difficult and politically contentious topic within the United States. As new districts are drawn, one difficulty in many states is the requirement to identify and maintain ‘communities of interest.’ This article explores the use of the mapper algorithm as a method for identifying clusters of similarities in demographic and economic census data. Using data from the American Community Survey, it presents the results found in Mecklenburg County, North Carolina, maps them geographically, and discusses their potential implications for communities of interest.

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1 Introduction

Legislative districting is a well known, and highly contentious, problem in American politics [10, 21]. While the exact methods and standards for drawing districts vary by state, according to the National Conference of State Legislators, twenty seven states include “preserving communities of interest” in their criteria for districting [25]. Much like variance across states with respect to methods, precise legislative definitions of communities of interest (COI) similarly vary or are left vague [15]. For example, Arizona states, “[d]istrict boundaries shall respect communities of interest to the extent practicable” [1], and Vermont districts must be consistent when practical with “recognition and maintenance of patterns of geography, social interaction, trade, political ties and common interests” [45]. One of the more precise definitions of a COI with respect to electoral districting can be found in California’s state constitution, which reads:

A community of interest is a contiguous population which shares common social and economic interests that should be included within a single district for purposes of its effective and fair representation. Examples of such shared interests are those common to an urban area, a rural area, an industrial area, or an agricultural area, and those common to areas in which the people share similar living standards, use the same transportation facilities, have similar work opportunities, or have access to the same media of communication relevant to the election process. Communities of interest shall not include relationships with political parties, incumbents, or political candidates [3].

In general, the interests that define a community may vary significantly, but broad areas can include demographic, economic, and geographic factors, amongst many others with the reasoning that such similarities will produce populations that share similar electoral and legislative interests [24]. While the COI requirements can be hard to interpret, and must be balanced with other requirements such as contiguity, compactness, and preserving political subdivisions, the requirement nonetheless provides a way to legally challenge a districting plan [33]. Further, they can become crucial components in how redistricting plans are created [16].

This paper presents the results of using mapper [39, 41] to find potential COI with American Community Survey (ACS) data, maps the clusters that emerge, and discusses both the implications of using said clusters as potential COI for purposes of districting and the implications and limitations of this approach moving forward. We note that while COI have been studied from geographic [14] and legal [20] perspectives, and mapper and persistent homology have been applied to topics like voting [13, 19] and gerrymandering [9], mapper remains a novel approach with respect to both COI and their impact upon districting. This paper tests the mapper algorithm’s potential as a new means for identifying historically disenfranchised and underrepresented populations that may warrant additional consideration or protection. Finally, this work was conducted as part of the Spatial Models and Electoral Districting NSF REU program in which a team of students from

mathematics, geography, and computer science utilized tools from computational topology, statistics, machine learning, and geography.¹

The paper is structured as follows. We first provide information about Mecklenburg County, North Carolina, and why it makes an ideal site for this approach. We then state our variables and why each is of interest, provide some background on our main tool, mapper [39], and discuss our parameter choices. We present three clusters of nodes that emerged through mapper, the characteristics that define these clusters, and how those characteristics may play into emergent COI. After comparing our methodology's results with those found through a traditional hierarchical clustering approach on the same data, we conclude the paper with a discussion of the need to validate our approaches through quantitative and/or qualitative social science methodologies for the detection and study of communities; these limitations inform both how this method may be used in other contexts as well as expanded upon in future work.

2 Mecklenburg County, North Carolina

Mecklenburg County, pictured in Fig. 1, has several unique characteristics that make it both a significant and appropriate subject for this research. It is the home of the city of Charlotte, the most populous city in the state. The county itself is a sanctuary county [4], and it has a diverse population along a number of identity markers [23]. Across the county, variegated, spatially distributed diversity is both a prime candidate for gerrymandering attempts and for the application of mapper for the detection of otherwise unconsidered COI. Mecklenburg County constitutes both the 9th and 12th North Carolina Congressional districts, the latter of which having figured prominently in high-profile United States Supreme Court cases on districting including *Easley v. Cromartie* [11], *Cooper v. Harris* [6], and most recently *Rucho v. Common Cause* [35].

While the Congressional district challenges have been much more visible nationally, the State Senate and House district plans have also been called into question recently. In 2015, a group of residents filed a complaint [7] challenging the district plans created after the 2010 Decennial Census on the grounds of being racially gerrymandered. A federal three-judge panel unanimously agreed with the challenge in 2016. In *North Carolina v. Covington* [29] the US Supreme Court ordered new maps be made. A combination of a 2017 state plan with alterations made by a special master were approved by the US Supreme Court and used in the 2018 election [30]. In 2019 a new district plan was enacted by the NC General

¹ The Co-Principal Investigator of the Spatial Models and Electoral Districting NSF REU site was a participant at the first Women in Computational Topology Workshop and the site itself is structured to offer opportunities and training to students traditionally underrepresented in STEM research, reflected by high participation percentages for women, gender-diverse individuals, and people of color.

Assembly through a bill later to become Session Law 2019-219 [38] was used for the 2020 election. We consider these three most recent State Senate district maps in our analysis. New maps will be drawn after the release of the 2020 Decennial Census data.

The North Carolina Constitution requires districts to have equal populations, be contiguous, and preserve political subdivisions [28]. Direct consideration of COI became an additional requirement for state legislative districts through the State Supreme court case [40], which stated such communities “should be considered in the formation of compact and contiguous electoral districts” (p.45 of [40]). Such language emphasises the inherent tension in identifying COI while satisfying other geographical considerations in electoral districting.

With respect to the history of contentious districting in North Carolina, its upcoming new districts, and its relative diversity, Mecklenburg County makes an ideal site to test mapper’s efficacy at identifying marginalized COI not well-recognized in existing districting approaches. Given the nature of census data and its historic tendency to undercount marginalized communities, some of which lack trust in government agencies [12, 17, 22, 36], mapper offers a particularly promising method for detecting and rendering legible such communities for districting purposes.

2.1 Data Selection

The 2017 House and Senate Plans Criteria for Redistricting Committees establishes 2010 federal Decennial Census data “as the sole basis of population for drawing legislative districts” [37]. Given the equal population requirement, legislative districts must ultimately be constructed out of census tabulation areas. The 2017

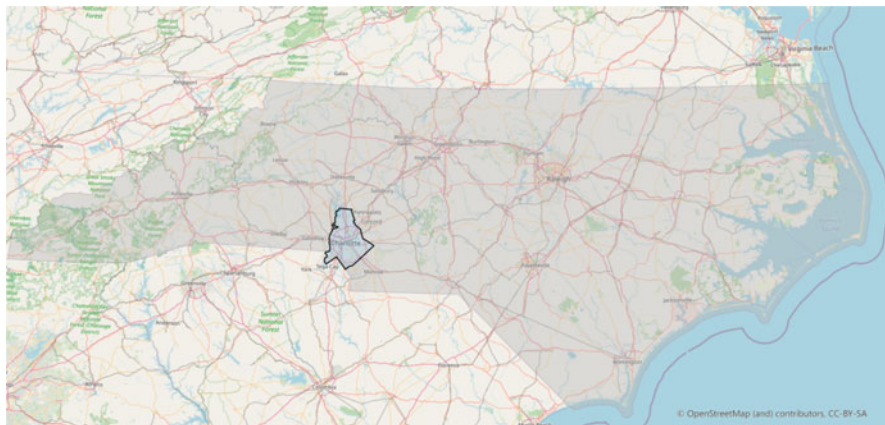


Fig. 1 The location of Mecklenburg County in North Carolina and the surrounding region

plans additionally allow for the consideration of election outcomes and explicitly forbid the consideration of any racial data. These restrictions apply to the drawing of districts and not to considerations of their appropriateness after the fact. As such, we had no such restrictions and were particularly interested in how mapper might identify emergent communities within demographic data that is known to severely undercount specific racial and socioeconomic populations [8, 17, 32].

The Census Bureau also collects information on education, employment, internet access, transportation, and other topics annually through the American Community Survey (ACS). This data is then used for the distribution of federal funding, for the placing of schools, hospitals, and highways, and other material outcomes that require the consultation of federal, standardized data [43]. This study only used ACS data for several reasons. First, the ACS provides both more up-to-date and detailed information than that provided by the 2010 Decennial Census. Second, using only open data allows for the easy reproduction and validation of our results [2]. Third, using federally created and maintained data both ensures that discussions on its validity are well documented (see, for example, [31] and [26]) and that the provenance for such data is well documented [42]. Finally, the known limitations and under-representations found in Census Bureau information provide a robust test for mapper as a means of detecting COI within socioeconomic and demographic population profiles.

Data for Mecklenburg County was obtained at the census block group level, the smallest areal unit at which the Census Bureau releases ACS data [44]. Given our interest in testing mapper’s ability to detect clusters in categorical data that may be incomplete, we chose 11 variables from the ACS 2018 5-year estimates related to race, education, and economic class which were then turned into proportions of households in the areal unit.

Table 1 lists the variables and reasoning for their selection. To find potential divisions in class, education, and age, we included proportion of households

Table 1 Variables selected from ACS 2018 5-year estimates

Variable (proportions of households)	Reason Chosen
No high school education	Educational divisions
Works in most popular industries	Occupational divisions
SNAP	Economic disadvantage
White collar occupations	Class divisions
Speaks English “less than well”	Immigrant communities
Households with minors	Families and communities with education and child services needs
Below 20% income	Economic disadvantage
Renting	Differences in renter and homeowner concerns, generational differences
Nonwhite	Lawsuits based on racial gerrymandering
Above 30% of income to rent	Economic disadvantage
Rural	Lawsuits based on gerrymandering

enrolled in the Supplemental Nutrition Assistance Program (SNAP), with income below the 20th percentile, that are renter occupied, with rent above 30% of income, with no high school diploma, with workers in the most common industries in Mecklenburg County, and with workers in white collar occupations. As the proportion of rented households has been increasing throughout the U.S., renting information can suggest generational and class differences that may constitute politically relevant communities and so related data was included. To help identify communities that may have family concerns such as education and child services, we considered proportion of households with children. As there are currently a number of growing immigrant communities in Mecklenburg County, we included proportion of households that speak English “less than well”. In *Cooper v. Harris* [6], the U.S. Supreme Court ruled that North Carolina had inappropriately used racial data in the construction of its 12th district and, as such, we included the proportion of households that are non-white. Finally, the proportion of households considered to be located in rural locations according to Census standards were included as rural and semi-rural locations were shown to be divided against their potential interests in *Common Cause v. Lewis* [5].

While geographic proximity can contribute to the idea of community, and neighborhoods can have similar legislative needs, we only considered the above demographic data for mapper and considered geographic proximity after the initial clustering. Each of these variables tracks historically with populations often at the center of districting challenges and with groups that are known to not be well-represented in official Census Bureau data [17]. Mapper’s ability to detect populations within this data would suggest not only possibilities for its use when working with categorical data, but its utility in districting challenges more generally. Here, it is important to note a distinction between the clusters that mapper detects and the communities that said clusters may signal; these differences and their implications are discussed in more detail in sections four and five.

3 Methods

The main approaches used for potential COI detection were mapper [39], hierarchical clustering, k-means, and singular value decomposition (SVD). For this project we used the Giotto-tda implementation, `gtda.mapper` [41]. Data processing and visualization were handled with QGIS, an open-source Geographic Information System. The choice to use open and available data, code, and software was intentional. Such choices offer reproducible approaches that are easier to interpret and enact into appropriate actions and policies [2].

3.1 Background on Mapper

We explore our demographic data using a topological data analysis (TDA) tool called the mapper algorithm, which was first proposed by Singh, et al. [39]. As tools from topology can be used to determine the shape of data, specifically when points are near others and when holes exist, TDA has been particularly successful for exploratory data analysis tasks such as clustering and feature extraction.

Mapper extracts 1-dimensional information from a data set and represents it by a graph. Nodes or vertices in the graph are collections of “similar” data points, and edges exist between nodes that share common data points. To do this, mapper starts with a point cloud X and a metric or other way to measure the distance between the points. It applies a filter function, $f : X \rightarrow \mathbb{R}^n$, for some small integer n (often 1 or 2 in order to lower the dimension of the data set), to the point cloud. It then selects a cover for the image in $\mathbb{R}^n$ (a finite collection of open sets in $\mathbb{R}^n$ such that every point in the image is in at least one of the sets), and finally clusters the collections of points that map to each set in the cover. The algorithm thus requires several choices as input:

1. A filter function $f : X \rightarrow \mathbb{R}^n$ —examples include projection onto one or more variable, eccentricity, and truncated SVD
2. Number of intervals to determine the covering
3. Overlap between sets in the cover—often between 10% and 50%
4. A clustering technique—examples include single linkage, complete linkage, and Ward hierarchical clustering

An example that helps visualize the process is provided in Fig. 2.

In the above example, we demonstrate the process with a filter function that maps to $\mathbb{R}^1$ for easier visualization. For this project, however, the filter function maps to $\mathbb{R}^2$ (we use the first two components from truncated SVD, as described below). While the process is similar, it worth noting how the cover of the image of the point cloud under the filter function changes. Specifically, the cover of the image is composed of rectangles created from intervals chosen over each component separately. The same number of intervals and the same proportion of overlap applies to each component. Figure 3 demonstrates this difference with an example of a possible cover of the image of a point cloud in $\mathbb{R}^2$. The points that are mapped to each rectangle are then clustered in the same way as the example above.

The focus of the mapper graph outputs tends to be on the flares and loops that emerge. These have been used to successfully identify groups of points that other clustering methods do not [18, 19, 27]. By construction, the nodes are not distinct clusters, data points can be assigned to more than one node. For example, in Fig. 2, we see two orange nodes resulting from clustering the points in the orange dashed box into a left group and a right group. Notice that there are points in both orange clusters that also fall within the red dotted box. As these points would also be included in the red node, we see that the red node and each of the orange nodes share members. The mapper graph indicates this by placing an edge between the overlapping nodes. As households may belong to more than one community, mapper

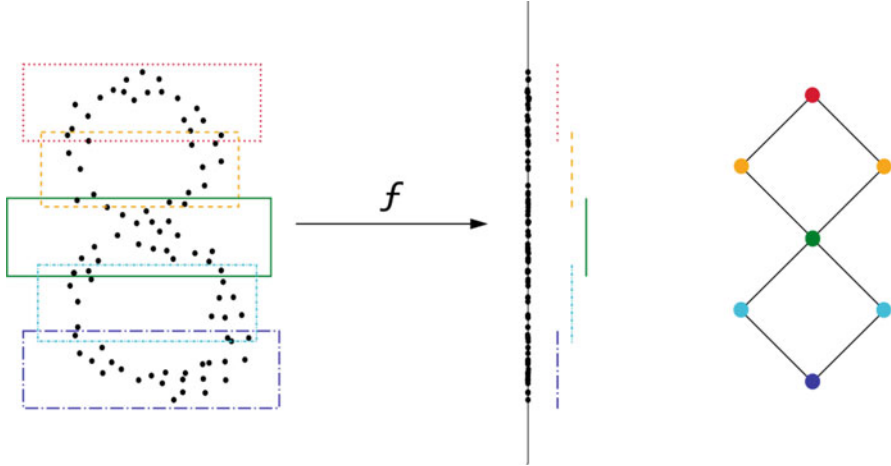


Fig. 2 An example of the mapper pipeline. The filter function projects the point cloud in the shape of a figure 8 to $\mathbb{R}$ by mapping a point to its height. A cover consisting of five intervals with 10% overlap is chosen. The points that map to each set in the cover are shown in the matching colored boxes over the original points. The points in each box are then clustered. The mapper graph on the right shows the resulting one-dimensional graph which captures the “holes” in the original data

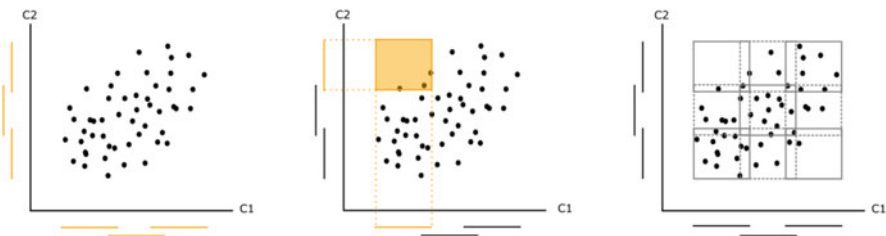


Fig. 3 Each component of the image is covered by three overlapping intervals. Each pair is then used to form a rectangle for the cover, resulting in nine total overlapping rectangles

has the potential to capture some of this multiplicity by placing households in more than one node, but some care needs to be taken when using the outputs to define a cluster, as we discuss in the results.

3.2 Mapper Inputs

For the filter function we chose truncated singular value decomposition (TSVD) to reduce the dimension of the data while keeping the most relevant variation information. After consideration of the variable distributions and the principal directions, we decided to use the first two components of TSVD on standardized data. While a larger number of components captures more variation in the data, the

extra dimensions in the image of the filter function produce more nodes and overlap than we found useful for our analysis.

The choice to use standardized data stemmed from comparisons of the first few principal components. While the original data and the standardized version both identified occupation, non-white, and rent vs. own as important axes, we found that the first principal component on the original data was over-weighting variables with larger internal variance. To address this, we used sklearn's `StandardScalar` function. Standardizing the data shifts some of the weight in the principal components from industry and rent as proportion of income to high school education, SNAP, and speaking English "less than well". Standardization also further emphasizes the rural variable. Details on the first two principal directions can be found in Table 3 in the Appendix. Our analysis here only includes the results from the standardized data, but we recognize that there may be some benefits to exploring the original data further.

For the cover, all of our mapper runs used a balanced cubical covering with an overlap fraction of 0.5 and with five intervals. While a smaller overlap could be explored further, initial tests demonstrated a preponderance of individual block groups appearing as distinct nodes—in other words, few nodes of a size viable for consideration as potential communities emerged at smaller overlaps. For the clustering method we chose First Simple Gap—Single Linkage. Taken together, the data and clustering approaches used produced sets of nodes suggestive of communities, which we now discuss.

4 Results

Mapper identified 18 nodes containing more than two block groups with these parameters and data; it is these nodes, and collections of them, that we examined as potential COI. A picture of the mapper graph is included in Fig. 4. The nodes are colored by where the block groups mapped along the first principal direction under TSVD. The darker colors roughly correspond to higher levels of white collar occupations in the more popular industries, and the lighter colors represent higher percentages of SNAP enrollments, no high school education, households under the 20th percentile in income, and non-white.

The size of a node corresponds to the number of block groups within that node. The smallest nodes all contain only one or two block groups, and most are already members of the larger nodes we have selected. Out of the 553 block groups analyzed, only eleven nodes (consisting of nine total block groups) were completely isolated without overlapping with larger nodes (they can be seen in the upper right of the mapper graph). The eight largest nodes (located in the center of the graph) each contained over 100 block groups, with two over 200. While these nodes are somewhat distinct from one another in terms of our demographic variables, due to the sheer size of each node, their over-arching characteristics weighted towards the mean demographics of the county as a whole, existing more as a summary of the

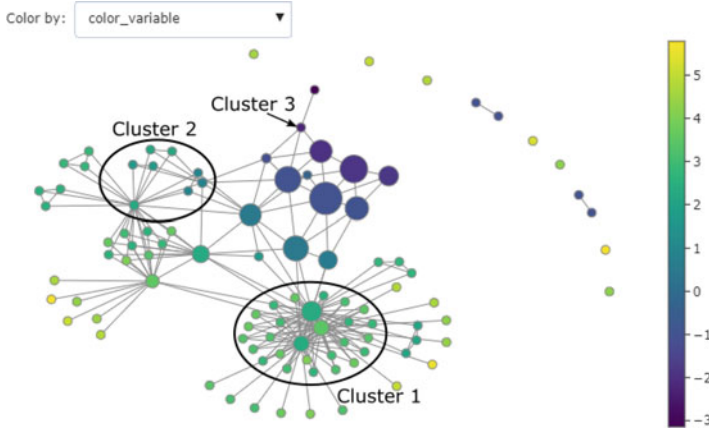


Fig. 4 The mapper output with clusters of interest circled. Node size indicates the number of block groups

county as a whole than of distinctions within it. Further, due to their size, these nodes geographically covered much of the county, making them not particularly useful for the analysis of potential COI.

Although we considered many potential groupings, we focus here on three smaller collections of nodes that stand out, describe what makes them distinct, discuss why these distinctions constitute potential COI, and visualize where they fall within current state legislative districts. The collections are circled and labelled in Fig. 4. When visualizing the larger clusters, we note the appearance of smaller, geographically proximate sub-groupings; it is with respect to these smaller, spatially proximate groupings that we examine current districting plans. These collections form communities in the sense of potentially having shared interests related to their demographic factors, but collectively they may not meet other definitions of community. For example, these clusters do not take into account contiguity or other spatial requirements for districting. Additionally, the demographic statistics analyzed by mapper cannot capture meaningful questions of community with respect to issues like internal social structure, daily lived experiences, and interpersonal interactions within and across community boundaries. This tension between what mapper reveals and the ‘ground truth’ of the potential communities speaks to the limits of purely quantitative approaches to social questions [34], a topic we return to in the conclusion.

4.1 The Clusters

Each cluster is described at a high level based on its demographic variable averages and visualized against the three sets of maps North Carolina has used for elections

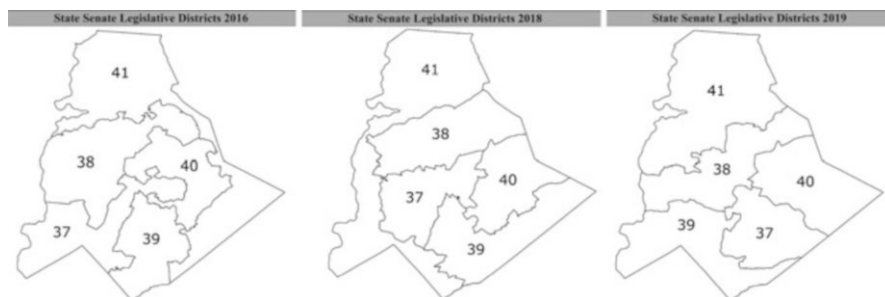


Fig. 5 The three district plans used most recently. The leftmost is from the 2010 Decennial Census, the middle was court-ordered and later approved in 2018, and the rightmost was enacted by the NC General Assembly in 2019

over the past decade using QGIS. The three districting plans are those described in Sect. 2: the enacted and challenged 2011 plans created after the last Decennial Census, the court-ordered plans approved in 2018, and the plans enacted through a NC General Assembly bill in 2019. The three districting plans are displayed in Fig. 5. We visualize the clusters detected by mapper and discuss their potential for forming COI against these three sets to illustrate how mapper performed against ongoing legislative issues with the drawing of districts in North Carolina. Specific variable averages and cluster sizes can be found in Table 2 in the Appendix. The terms cluster and potential community of interest are used intentionally and carefully to demarcate what the mapper output has detected with what this may implicate within the existing communities and households of Mecklenburg county.

For reference while discussing our results, in our 2018 ACS data Mecklenburg County was estimated to have around 1.05 million residents, and each of the five districts is required to contain within 5% of an even division of the population [40]. This puts the population of each district at roughly 210,000 (the 2011 districting plan has each district around 184,000). Additionally, Charlotte, the largest city in Mecklenburg County, sits at the center of the county geographically.

Cluster 1—Lower Education and Wealth

Cluster 1 (C1) consists of 135 block groups with a population of roughly 226 thousand, and is comprised of three large nodes at the bottom of our mapper graph (the small surrounding nodes are all subsets of these three). It is characterized by lower levels of wealth, education, and homeownership. Compared to the Mecklenburg County averages, this cluster has a much higher proportion of households enrolled in SNAP, having an income below the 20th percentile, and renting, with more than half of the households spending more than 30% of their income on rent. While these indicate less wealth, high proportions of households renting may indicate younger populations. C1 also has a racial make-up which is different from the rest of Mecklenburg with the majority of the cluster being non-white. Many more households are without a high school diploma and few households have occupants working in white collar positions.

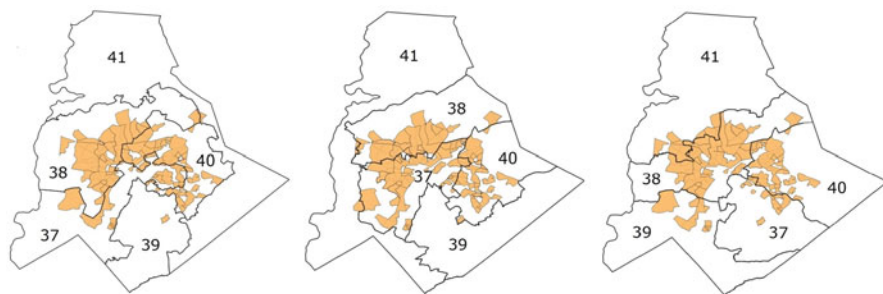


Fig. 6 The collection of block groups within C1 mapped against three recent districting plans. The 2011 enacted plan is in the upper left, the 2018 court-ordered plan is in the upper right, and the 2019 enacted plan is on the bottom

Geographically, as can be seen in Fig. 6, this cluster is mostly contiguous with a few isolated block groups near the edges. It appears to be a geographic cluster as well as a group with common demographic factors. The block groups predominantly sit within Districts 37, 38, and 40 in the 2011 enacted plan with $\sim 40\%$ of the population in District 40, $\sim 35\%$ into District 38, and $\sim 20\%$ in District 37. In the 2019 enacted plan, the proportions shift to $\sim 31\%$ in District 40, $\sim 39\%$ in District 38, and roughly 10% in each of the other three. While this has C1 as close to 50% of the total population of District 40 in the 2011 plan and close to 50% of total the population of District 38 in the 2019 plan, given that the population numbers are estimates and the voting population specifically has not been taken into account, it is unclear whether or not C1 has the majority in either case. While this hints at the districting being racially gerrymandered in both cases, more careful analysis needs to be done.

We note that the newer 2019 plan disperses the block groups outside of Districts 38 and 40 somewhat equally, causing some of the more isolated block groups in the 2011 plan to be combined with other C1 block groups. On the other hand, many blocks in the northwest region have been shifted into District 41, which generally contains more affluent nodes, and they are now separated from the more central cluster. Many of the block groups in Districts 37 and 39 have been swapped just due to the substantial shift in lines. The isolated block group near the south in District 39 in the 2011 plan became a part of District 37 in 2019, joining other members of C1.

As the total population in this cluster constitutes over 20% of the county's total population, it is too large to be a single district. Further, putting the vast majority in a single district might constitute packing, an established form of gerrymandering. On the other hand, by splitting the cluster with its shared demographic and geographic ties up over all five of the districts, no potential grouping may emerge as the majority in a single district. While it is unclear if this cluster constitutes a community with shared interests that could be met from being able to elect a candidate of choice,

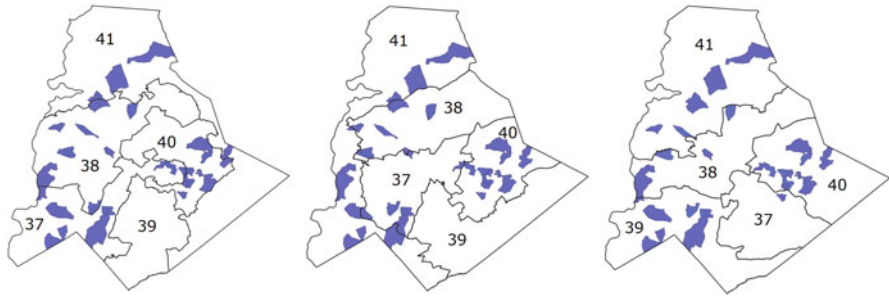


Fig. 7 The collection of block groups within C2 mapped against three recent districting plans. The 2011 enacted plan is in the upper left, the 2018 court-ordered plan is in the upper right, and the 2019 enacted plan is on the bottom

it is possible that a potential community is not being represented, and that racial characteristics may be being used to draw districts.

Cluster 2—Immigrants with Children

Cluster 2 (C2) is made up of two nodes toward the left of the mapper plot and contains 36 block groups with just over 84 thousand in population. This cluster has unique characteristics with regards to the proportion of households that speak English “less than well” and have minors in them as compared with the rest of Mecklenburg county. C2 is also different from the rest of Mecklenburg when it comes to occupation and industry type. The composition of common industries in this community is much lower than average. Similarly this cluster has a much lower proportion of workers categorized as white collar.

C2 residents potentially share a number of social interests. Compared to Mecklenburg as a whole, C2 has a higher percentage of households with minors which may indicate shared interests with regards to childcare and education opportunities. Additionally, C2 has a higher proportion of people who speak English “less than well”, and over half of the households are non-white—both indicators of a larger than average immigrant population for the county. Along other axes measured, C2 aligns roughly with the averages of Mecklenburg county.

Contrasting the maps in Fig. 7 of this cluster with Cluster 1, we see that C2 is spread out throughout the county with almost no block groups contiguous with one another. This suggests that while this cluster is demographically similar along the axes analyzed, members may in fact be part of very distinct individual communities. The 2019 districting plan does seem to capture most of the collection of block groups to the east in District 40 and the block groups in the southwest region in District 39 (which had no block groups from this community in it in the 2010 Census plan).

Cluster 3—Wealthy Homeowners with Children

Cluster 3 (C3) contains one node and consists of 21 block groups and a population of just over 55 thousand. The characteristics that make C3 unique are that it is significantly more rural than average for Mecklenburg with over 75% of the house-

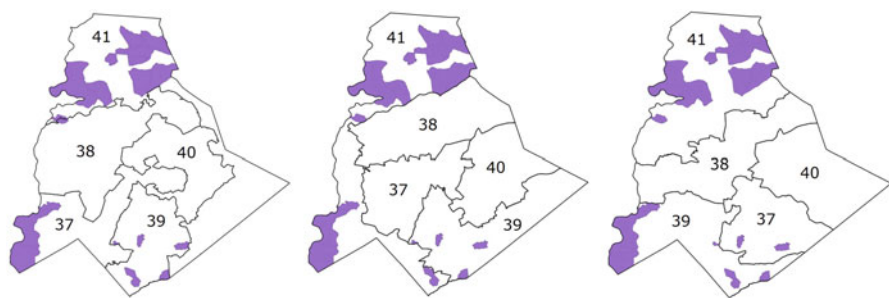


Fig. 8 The collection of block groups within C3 mapped against three recent districting plans. The 2011 enacted plan is in the upper left, the 2018 court-ordered plan is in the upper right, and the 2019 enacted plan is on the bottom

holds working in white collar positions. Additionally, members of the cluster have high rates of minors in the household and of home ownership. C3 is predominantly wealthy, white, and educated with low proportions of the households below the 20th income percentile and enrolled in SNAP.

Geographically, as can be seen in Fig. 8, C3 is not wholly contiguous, but consists of two main groupings of block groups in the north and southwest of Mecklenburg, and a few additional block groups to the south. All members are located far from the Charlotte city center. The 2011 districting plan places members of this cluster in four of the five districts, the court-ordered 2018 plan has them in only two (where the north and southwest groupings are joined in a single district), and the 2019 plan divides them into three. Given the dispersed nature of this cluster, it is highly unlikely to find them represented by a single congressional district.

Taken together, these three clusters suggest that mapper can detect potential communities along shared demographic and economic categories. Of particular interest is its ability to detect potentially shared racial groups and how they have been split by districting maps found to be racially gerrymandered. However, what remains to be seen is whether mapper's clusters are unique to its approach, and therefore offer something new for consideration, or are simply detecting identical clusters as to those that might be found through other techniques. In the next section, we compare mapper's results to those found through hierarchical clustering of the same data.

4.2 Hierarchical Clustering Comparison

To compare our mapper results with traditional hierarchical clustering methods, we ran our data through single linkage hierarchical clustering using the `hclust` function from R stats package. This method was chosen specifically as single linkage clustering was used as the clustering technique within mapper, the goal being to

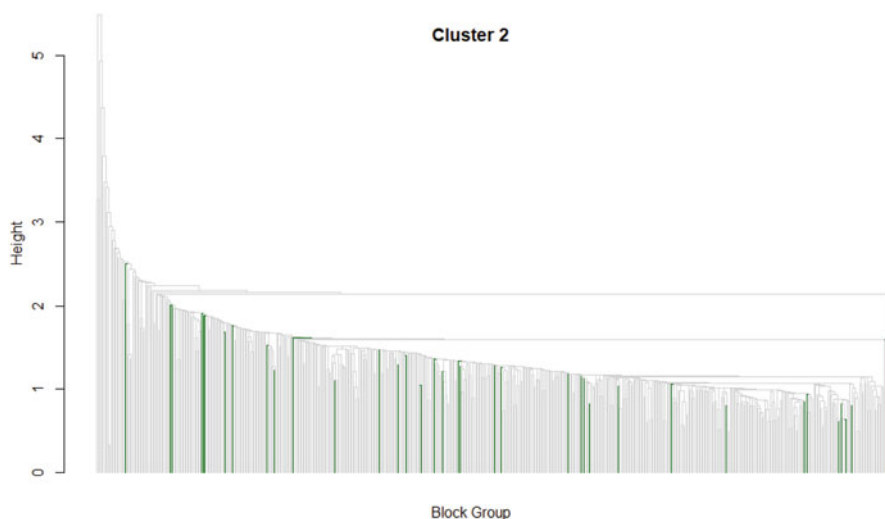


Fig. 9 The dendrogram here shows the block groups from Cluster 2 in green. The spread out nature of the block groups indicates that single linkage hierarchical clustering does not identify Cluster 2 as a collection of similar block groups

demonstrate that mapper identified collections of block groups that hierarchical clustering did not identify when used as the only method.

Hierarchical clustering starts with each block group being a single cluster, then iteratively identifies the two closest clusters and merges them. This process is continued until all clusters are merged into a single cluster. To obtain a set of clusters, the process can be stopped once the desired number of clusters has been reached.

The clustering results are often displayed in a dendrogram, which places nearby clusters next to each other and shows two merged clusters with a horizontal bar connecting them. Each vertical line is an individual block group and the height of the horizontal bar connecting block group clusters corresponds to the distance between the two clusters. The results of the hierarchical clustering are displayed in Fig. 9. In this figure we have colored the block groups in Cluster 2 green using the dendextend R package to see where they fall within the hierarchical clustering. One can see that the lines are scattered throughout, rather than being grouped nearby, demonstrating that single linkage hierarchical clustering does not seem to identify the members of Cluster 2 as particularly similar. This further suggests that mapper offers potential new avenues for the examination of socioeconomic and demographic data.

Similar results are obtained for Clusters 1 and 3. The dendrograms with their block groups colored in green are available in the appendix.

5 Conclusions and Next Steps

In this article, we presented the results of using the mapper algorithm to analyze ACS demographic and economic data for Mecklenburg County, North Carolina. Three detected clusters were of particular interest as they presented clear similarities and similar clusters were not detected by hierarchical clustering. This suggests that mapper has unique capabilities in detecting similarities within sociological data that other methods would not.

Preserving COI remains a difficult problem in districting—both in terms of the precise definitions of COI and in their detection once defined. It is important to note that in this article we presented clusters of households that share similar profiles with respect to other county residents; however, whether these clusters represent COI as defined in a legal or sociological sense remains to be tested. Further, even if many of these clusters do represent COI appropriate for influencing districting decisions, given their non-contiguous nature that may be extremely difficult.

Mapper, then, offers an important tool for the analysis of demographic and economic data; but, it is not a definitive one. Clustering algorithms cannot solve districting questions, but they can identify important topics for further consideration. For example, Cluster 1's segmentation into five districts in both the 2011 enacted and 2019 enacted districting plans adds further evidence of racially motivated gerrymandering. This simultaneously suggests that mapper holds promise as a tool for detecting this issue in other circumstances and that 'ground truth' validation of this approach is necessary. We see mapper as one tool amongst many for the detection and analysis of potential communities, highlighting the need for scholars and activists to work across disciplinary boundaries—computational topology, geography, sociology, etc.—on important, otherwise intractable topics such as districting.

Acknowledgments This project was funded through the NSF REU Site: Spatial Models and Electoral Districting (Award #1851696). We would especially like to thank the first summer REU cohort for their work that set up later projects including this one. We would additionally like to thank the reviewers for their time and feedback improving this manuscript.

Appendix

Table 2 includes the averages of the block group proportions of households for each selected variable and Table 3 provides the first and second principal directions from TSVD. Figure 10 provides dendrograms from single linkage hierarchical clustering with Clusters 1 and 3 members identified in green. Access to the original data set can be obtained from the authors upon request.

Table 2 Averages of the block group proportions of households data by cluster identified (from 2018 ACS 5-year)

Variable	All BG	C1	C2	C3
No high school education	0.0796	0.1562	0.1073	0.0279
Works in most popular industries	0.6687	0.6328	0.5082	0.6513
SNAP	0.1157	0.2597	0.1406	0.0184
White collar occupations	0.6615	0.4610	0.4372	0.8430
Speaks English “less than well”	0.0428	0.0708	0.1497	0.0080
Households with minors	0.3137	0.3009	0.4152	0.4767
Below 20% income	0.1792	0.3359	0.1642	0.0491
Renting	0.4245	0.6353	0.4177	0.1023
Nonwhite	0.4447	0.7427	0.5716	0.2212
Above 30% of income to rent	0.4447	0.5666	0.4787	0.2617
Rural	0.0103	0.0000	0.0186	0.1290
Number of Block Groups	553	135	36	21

Table 3 Principal directions from TSVD

Variable	1st vector	2nd vector
No high school education	0.343	−0.006
Works in most popular industries	−0.198	−0.480
SNAP	0.401	−0.136
White collar occupations	−0.415	−0.182
Speaks English “less than well”	0.257	0.408
Households with minors	0.017	0.559
Below 20% income	0.396	−0.238
Renting	0.307	−0.286
Non-white	0.376	−0.052
Above 30% of income to rent	0.222	−0.047
Rural	−0.053	0.310

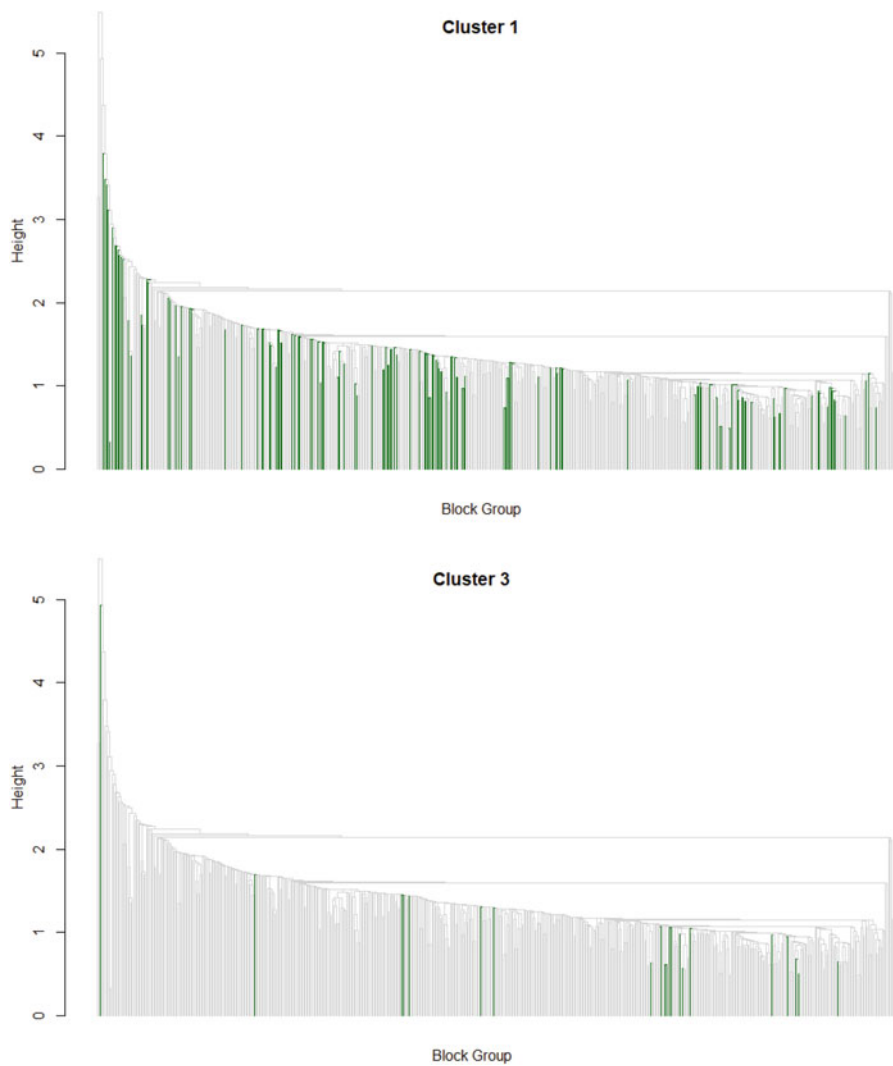


Fig. 10 The dendrograms here show the block groups from Clusters 1 and Clusters 3 in green, respectively. The block groups being spread out throughout indicates that hierarchical clustering does not identify the same clusters

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